

JEE-Main-29-01-2025 (Memory Based)
[MORNING SHIFT]

Physics

Question: Assertion: At the peak of the mountain, the time period of pendulum increases.

Reason: Time period of pendulum increases with decrease in g .

Options:

- (a) Assertion is correct, Reason is incorrect
- (b) Assertion is incorrect, Reason is correct
- (c) Assertion is incorrect, Reason is incorrect
- (d) Assertion is correct, Reason is correct

Answer: (c)

Question: The velocity of a particle moving on a straight line varies with time as

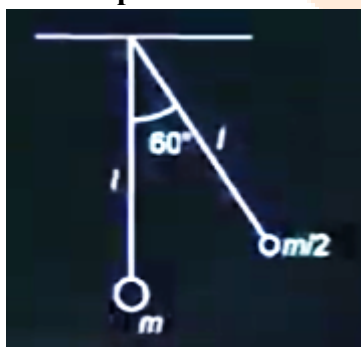
$$v = At^2 + \frac{Bt}{C + t}, \text{ where } A, B, C, \text{ are constants. Find the dimension of } ABC.$$

Options:

- (a) L^2T^{-2}
- (b) L^2T^{-1}
- (c) L^2T^{-3}
- (d) LT^{-3}

Answer: (c)

Question: A pendulum of mass $m/2$ is released from a given situation. Find speed of another pendulum after collision. ($e = 1$)



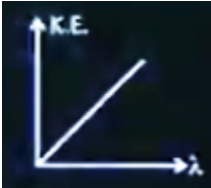
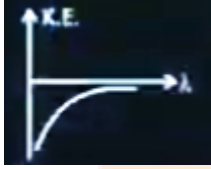
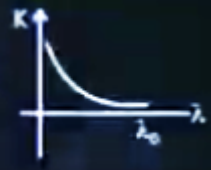
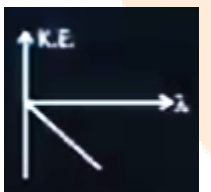
Options:

- (a) $\sqrt{\frac{3}{2}gl}$

- (b) $\frac{2}{3} \sqrt{gl}$
 (c) $\sqrt{\frac{gl}{3}}$
 (d) $\frac{1}{3} \sqrt{gl}$

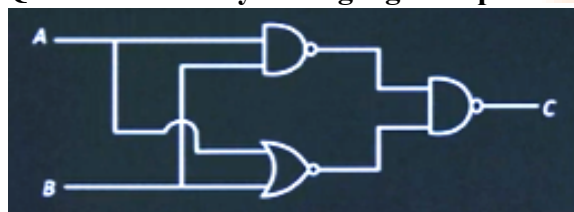
Answer: (b)

Question: The graph between wavelengths (7.) of incident light and kinetic energy (K.E.) of photoelectrons in photoelectric effect is
Options:

- (a) 
 (b) 
 (c) 
 (d) 

Answer: (c)

Question: Identify the logic gate represented by the circuit shown below.



Options:

- (a) OR Gate
 (b) NAND Gate
 (c) AND Gate
 (d) NOR Gate

Answer: (a)

Question: Statement-1: Electromagnetic wave have both energy and momentum.

Statement-2: Rest mass of photon is zero.

Options:

- (a) Statement-1 is correct, statement-2 is correct
- (b) Statement-1 is correct, statement-2 is incorrect
- (c) Statement-1 is incorrect, statement-2 is correct
- (d) Statement-1 is incorrect, statement-2 is incorrect

Answer: (a)

Question: Two projectiles were launched from same position simultaneously only same speed one of the projectile was launched at angle $(45 - \alpha)^\circ$ and the other at an angle of $(45 + \alpha)^\circ$. Find the ratio of maximum height of the projectile.

Options:

- (a) $\frac{1 - \sin \alpha}{1 + \sin \alpha}$
- (b) $\frac{1 + \sin 2\alpha}{1 - \sin 2\alpha}$
- (c) $\frac{1 + \sin \alpha}{1 - \tan \alpha}$
- (d) $\frac{1 + \tan \alpha}{1 - \cos \alpha}$

Answer: (b)

Question: A river is flowing with speed 9 km/h. Boat is going downstream-speed of boat in still water is 27 km/h. A person in boat throws a ball upwards with speed 10 m/s. Find range of the ball as seen by an observer at bank of river

Options:

- (a) 10 m
- (b) 20 m
- (c) 25 m
- (d) $20\sqrt{3}$ m

Answer: (b)

Question: Which of two physical quantities have same dimensions?

Options:

- (a) Angular momentum and Planck's constant
- (b) Torque and moment of inertia
- (c) Impulse and surface tension
- (d) Momentum and work done

Answer: (a)

Question: If radius of first Bohr's orbit of H-atom is a . Then find the radius of 2nd Bohr's orbit of H-atom.

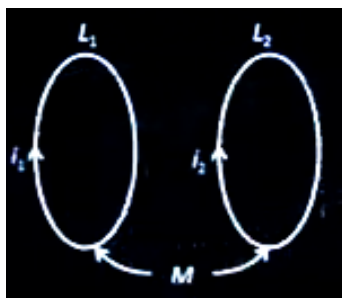
Options:

- (a) $8a_0$

- (b) $4a_0$
- (c) $2a_0$
- (d) $6\pi a_0$

Answer: (b)

Question: Two coils having self-inductance L_1 and L_2 are placed closely such that they have a mutual inductance M . If they carry currents i_1 and i_2 as shown in the figure then the induced emf in coil 1 is



Options:

- (a) $-L_1 \left(\frac{di_1}{dt} \right) + M \left(\frac{di_2}{dt} \right)$
- (b) $-L_1 \left(\frac{di_1}{dt} \right) - M \left(\frac{di_2}{dt} \right)$
- (c) $-L_1 \left(\frac{di_2}{dt} \right) + M \left(\frac{di_1}{dt} \right)$
- (d) $-L_1 \left(\frac{di_2}{dt} \right) - M \left(\frac{di_1}{dt} \right)$

Answer: (c)

Question: When a ball is kept under sea at depth 2.5 km. Find percentage change in its volume. If bulk modulus of water is 2×10^9 Pa.

Options:

- (a) 2%
- (b) 1.5%
- (c) 1.25%
- (d) 2.75%

Answer: (c)

Question: Assertion: A negative potential is required to stop the photoelectron.

Reason : Speed of electron decreases when a negative potential is applied in a photo cell.

Options:

- (a) Assertion is correct but Reason is false
- (b) Assertion is correct and Reason is also correct
- (c) Assertion is false but Reason is correct
- (d) Assertion is false and Reason is also false

Answer: (b)

Question: If electric dipole of dipole moment $\vec{P}$ is placed in electric field $\vec{E}$ with $\vec{P} \parallel \vec{E}$. It is rotated slightly (and slowly) and released. Find the time period of oscillation of dipole (moment of inertia of dipole is I).

Options:

(a) $T = 2\pi\sqrt{\frac{I}{PE}}$

(b) $T = \frac{1}{2\pi}\sqrt{\frac{PE}{I}}$

(c) $T = 2\pi\sqrt{\frac{PE}{I}}$

(d) $T = \frac{1}{2\pi}\sqrt{\frac{PI}{E}}$

Answer: (a)

Question: In adiabatic process of closed system, work done by the gas depends explicitly on

Options:

(a) Change in volume

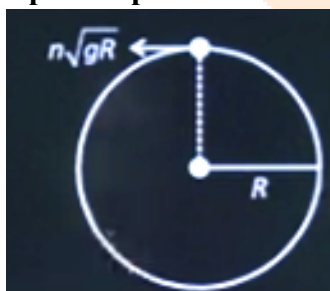
(b) Change in pressure

(c) Change in temperature

(d) Change in number of moles

Answer: (c)

Question: A particle is able to complete the vertical circular motion with speed $n\sqrt{gR}$ at top-most point. Find the ratio of- $\frac{KE_{(Bottom)}}{KE_{(Top)}}$



Options:

(a) $\frac{n^2 + 4}{n}$

(b) $\frac{n^2 + 4}{n}$

$$(c) \frac{n^2 + 2}{n^2 + 4}$$

$$(d) \frac{n^2}{n^2}$$

Answer: (d)

Question: Match the correct option for List-I and List-II, where symbols have usual meanings.

	List-I		List-II
(A)	Electric field inside the spherical shell	(i)	$\frac{\sigma}{2\epsilon_0}$
(B)	Electric field just outside the spherical shell	(ii)	$\frac{\sigma}{\epsilon_0}$
(C)	Electric field inside the charged parallel plate capacitor	(iii)	Zero
(D)	Electric field of infinite charge sheet	(iv)	$\frac{2\sigma}{\epsilon_0}$

Options:

(a) A-(iii), B-(ii), C-(iv), D-(ii)

(b) A-(iii), B-(ii), C-(ii), D-(i)

(c) A-(iii), B-(ii), C-(ii), D-(iv)

(d) A-(iv), B-(iii), C-(i), D-(ii)

Answer: (b)

Question: In a hydraulic lift, the two sides have areas $A_1 = 25 \text{ cm}^2$ and $A_2 = 100 \text{ cm}^2$. If a force of 100 N is applied normally on the area A_1 , then the force on the area A_2 is _____ N.

Options:

(a) 500 N

(b) 600 N

(c) 400 N

(d) 450 N

Answer: (c)

Question: Find magnitude of component of torque about origin in z-direction when

force $\vec{F} = \hat{i} - \hat{j} + \hat{k}$ acts at (1, 1, 1).

Options:

(a) -2

(b) 2

(c) 3

(d) -3

Answer: (a)



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[MORNING SHIFT]
Chemistry

Question: Which of the following is animal starch?

Options:

- (a) Glycogen
- (b) Lactose
- (c) Amylopectin
- (d) Amylose

Answer: (a)

Question: Statement-1: Correct order of ionic radius for Mg^{2+} , Na^+ , O^{2-} & F^- is $\text{F}^- > \text{O}^{2-} > \text{Na}^+ > \text{Mg}^{2+}$

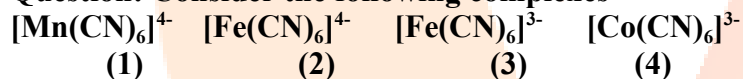
Statement-2: Correct order of magnitude of gain Enthalpy for 17th group follows order $\text{Cl} > \text{F} > \text{Br} > \text{I}$ (Magnitude only)

Options:

- (a) Both Statement-1 & Statement-2 are correct
- (b) Statement-1 is correct & statement-2 is correct
- (c) Both Statement-1 & Statement-2 are incorrect
- (d) Statement-1 is incorrect & statement-2 is correct

Answer: (d)

Question: Consider the following complexes



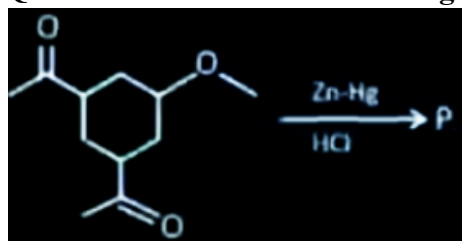
Correct order of CFSE (Δ_0) will be

Options:

- (a) $3 > 4 > 2 > 1$
- (b) $4 > 3 > 2 > 1$
- (c) $4 > 3 > 1 > 2$
- (d) $3 > 4 > 1 > 2$

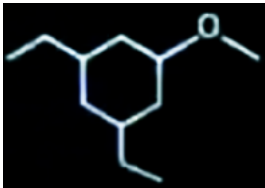
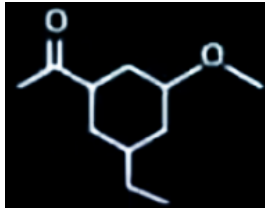
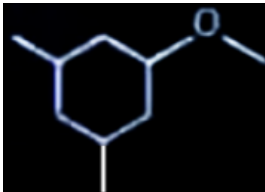
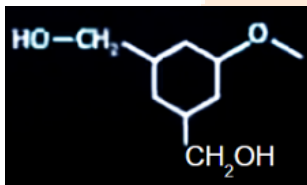
Answer: (b)

Question: Consider the following reaction



Identify the final product P.

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (a)

Question: What is the value of van't Hoff Factor for A_2B is 30% of A_2B is dissociated?

Options:

- (a) 1.60
(b) 1.30
(c) 1.50
(d) 1.20

Answer: (a)

Question: Find the order of the reaction $A + B \rightarrow F$

If the mechanism of the reaction is given below:

Step-1: $A + B \rightarrow D$ (slow)

Step-2: $D \rightarrow C + E$ (fast)

Step-3: $C + E \rightarrow F$ (fast)

Options:

- (a) 1
(b) 3
(c) 2
(d) 4

Answer: (c)

Question: Match the following List-I with List-II

	List-I		List-II
A	$[\text{Co}(\text{OH})_3]^{3-}$	(i)	sp^3d^2

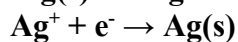
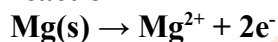
B	$[\text{FeF}_6]^{3-}$	(ii)	d^2sp^3
C	$[\text{Ni}(\text{CO})_4]$	(iii)	dsp^2
D	$[\text{PtCl}_4]^{2-}$	(iv)	sp^3

Options:

- (a) A-i, B-ii, C-iii, D-iv
- (b) A-ii, B-i, C-iii, D-iv
- (c) A-i, B-ii, C-iv, D-iii
- (d) A-ii, B-i, C-iv, D-iii

Answer: (d)

Question: What is the correct Nearest equation representation for the following cell reaction



Options:

- (a) $E_{\text{cell}} = E_{\text{cell}}^o - \frac{RT}{nF} \ln \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]^2}$
- (b) $E_{\text{cell}} = E_{\text{cell}}^o - \frac{RT}{nF} \ln \frac{[\text{Ag}^+]^2}{[\text{Mg}^{2+}]}$
- (c) $E_{\text{cell}} = E_{\text{cell}}^o - \frac{RT}{nF} \ln \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]}$
- (d) $E_{\text{cell}} = E_{\text{cell}}^o - \frac{RT}{nF} \ln \frac{[\text{Ag}^{2+}]}{[\text{Mg}^{2+}]}$

Answer: (a)

Question: The correct order of melting point of d-block elements is:

Options:

- (a) $\text{Fe} > \text{Mn}$
- (b) $\text{Tc} > \text{Ru}$
- (c) $\text{Os} > \text{Re}$
- (d) $\text{Ta} > \text{W}$

Answer: (a)

Question: Λ_m is directly proportional to $\sqrt{c}$ for an electrolyte, then molar conductance for the same electrolyte at infinite dilution shows

Options:

- (a) Small increase
- (b) Small decrease
- (c) Sharp increase
- (d) Sharp decrease

Answer: (a)

Question: Given ionisation enthalpy of element E(g) is 300 kJ/mol and electron gain enthalpy of A, B, C and D gases atoms are -320 kJ/mol, -200 kJ/mol and -250 kJ/mol, then what will be the correct order of ionic nature of compounds?

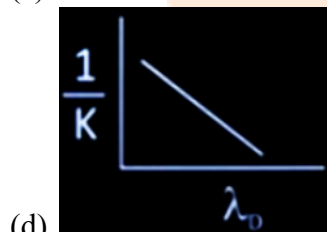
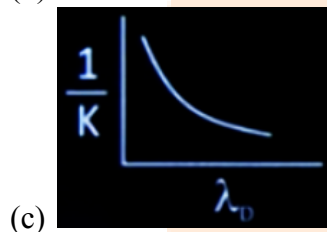
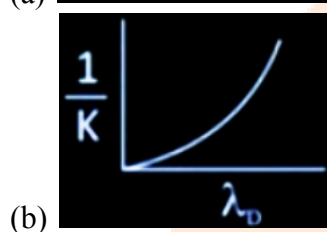
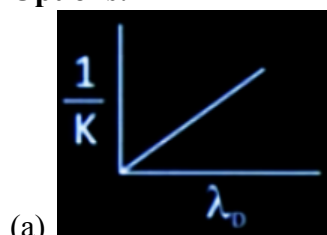
Options:

- (a) $EB > EA > ED > EC$
- (b) $EB > EA > EC > ED$
- (c) $EC > ED > EA > EB$
- (d) $EC > ED > EB > EA$

Answer: (a)

Question: Graph between deBroglie wavelength (λ_D) and kinetic energy (K) of an electron is

Options:



Answer: (b)

Question:

Which of the following ions is strongest oxidising agent

Given:

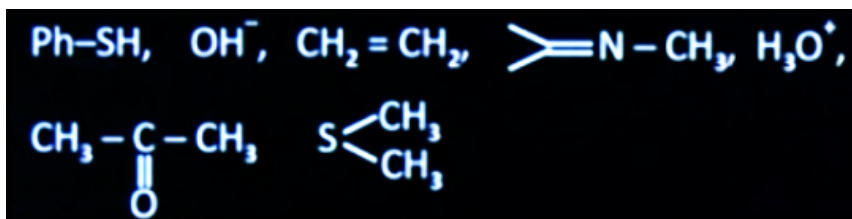
$$E_{Al^{3+}/Al}^o = -2.7V \quad E_{Cu^{2+}/Cu}^o = 0.34V \quad E_{Pb^{4+}/Pb^{2+}}^o = 1.8V \quad E_{Ti^{3+}/Ti}^o = -1.21V$$

Options:

- (a) Al^{3+}
- (b) Cu^{2+}
- (c) Pb^{4+}
- (d) Ti^{3+}

Answer: (c)

Question: How many of following can act as nucleophile

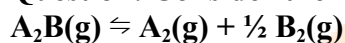


Options:

- (a) 5
- (b) 6
- (c) 7
- (d) 4

Answer: (b)

Question: Consider the following reaction



If P is total pressure at equilibrium & K_p is equilibrium constant. Then α in terms of K_p & P is (Assume $\alpha \ll 1$)

Options:

- (a) $\sqrt{\frac{K_p}{P}}$
- (b) $\sqrt[4]{\frac{K_p}{P}}$
- (c) $\sqrt{\frac{2K_p}{P}}$
- (d) $\sqrt[3]{\frac{2K_p^3}{P}}$

Answer: (d)

Question: Calculate the total number of sigma and π bond in the given molecule?

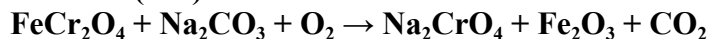


Options:

Answer: $(11 + 4 = 15)$

Question: Chromite ore + $\text{Na}_2\text{CO}_3 + \text{O}_2 \rightarrow$ insoluble product containing Fe Calculate the molar mass of insoluble product formed. (Given : Molar mass of Cr = 52 g/mol, Na = 23 g/mol, Fe = 56 g/mol, O = 16 g/mol)

Answer: (160)

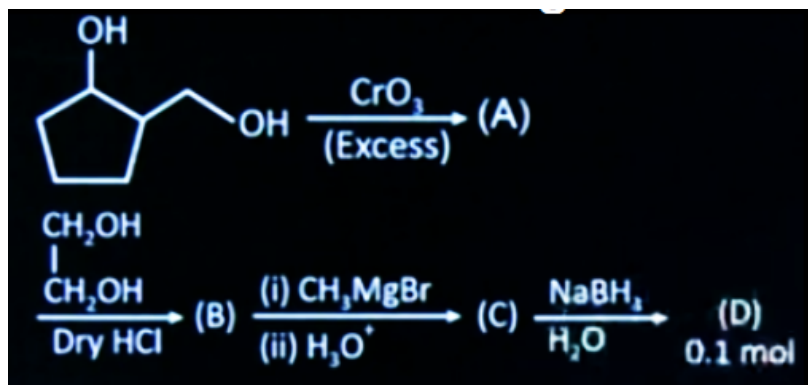


$$M_w = 2 \times 56 + 3 \times 16$$

$$= 112 + 48$$

$$= 160$$

Question: Consider the following reaction



Find the mass of final product (D) formed in gm(nearest integer)

Answer: (13)

JEE-Main-29-01-2025 (Memory Based)
[MORNING SHIFT]
Maths

$$80 \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$$

Question:

Options:

- (a) $2 \log 5$
- (b) $8 \log 3$
- (c) $3 \log 8$
- (d) $2 \log 3$

Answer: (b)

Question:

$\vec{a} = 2i - j + 3k, \vec{b} = 3i - 5j + k$, if $\vec{a} \times \vec{c} = \vec{c} \times \vec{b}$ and $(\vec{a} + \vec{c}) \cdot (\vec{b} + \vec{c}) = 168$ then find maximum value of $|\vec{c}|^2 =$

Options:

- (a) 154
- (b) 308
- (c) 300
- (d) 77

Answer: (b)

Question: $L_1 = \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$, $L_2: \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$. Let the line L_3 passes through the point (α, β, γ) perpendicular to L_1 & L_2 and L_3 intersect line L_1 then $|5\alpha - 11\beta - 8\gamma|$.

Options:

- (a) 25
- (b) 18
- (c) 16
- (d) 20

Answer: (a)

Question: $|z_1 - 8 - 2i| \leq 1$ and $|z_2 - 6 + 8i| \leq 2$ then minimum value of $|z_1 - z_2|$ is equal to

Options:

- (a) $2\sqrt{25} + 3$
- (b) $4\sqrt{21} + 5$
- (c) $2\sqrt{26} - 3$
- (d) $4\sqrt{21} - 5$

Answer: (c)

Question: The minimum value of n for which the number of integer terms in the binomial expansion of $(7^{1/3} + 11^{1/2})^n$ is 183, is

Answer: (2184)

Question: In an A.P. $S_3 = 54$ and S_{20} lies between 1600 and 1800 and if the common difference in this A.P. is an integer. Then find 11th term of this A.P.

Options:

- (a) 69
- (b) 72
- (c) 45
- (d) 90

Answer: (d)

Question: $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} =$

Options:

- (a) $\frac{1}{5}$
- (b) $\frac{2}{3}$
- (c) $\frac{5}{3}$
- (d) $\frac{1}{3}$

Answer: (c)

Question: Area bounded by the curves $2y \geq x^2 + 3$, $y \geq |x-1|$ & $y + |x| \leq 3$ is A then find the value of 6A

Options:

- (a) 10
- (b) 12
- (c) 7
- (d) 14

Answer: (a)

Question: Number of 7 digit numbers that can be made using the digits 1, 2, 3 such that sum of digits is 11?

Answer: (161)

Question: Let R be a relation defined on $(0, \frac{\pi}{2})$ such that $xRy \Leftrightarrow \sec^2 x - \tan^2 y = 1$ then

Relation R is

Options:

- (a) Equivalence
- (b) Reflexive & transitive only
- (c) Symmetric & transitive only
- (d) Neither reflexive nor transitive

Answer: (a)

Question: The minimum value of p such that

$$\lim_{x \rightarrow 0} \left(\left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \dots + \left\lfloor \frac{p}{x} \right\rfloor \right) - x^2 \left(\left\lfloor \frac{1}{x^2} \right\rfloor + \left\lfloor \frac{2}{x^2} \right\rfloor + \dots + \left\lfloor \frac{9}{x^2} \right\rfloor \right) \geq 1, \text{ is equal to}$$

Answer: (10)